

Aggregating Judgments When People Receive Advice from an Artificial Intelligence System: Baseline Study Postmortem

Forecasting Method	Mean Absolute Error (MAE)
Crowd's average initial forecast, $\bar{f}^0$	6.46
Crowd's average final forecast, $\bar{f}^1$	5.71
AI advice, f^a	5.64
$\hat{\theta}_\lambda$ obtained using the judgment aggregation procedure proposed in the preregistration	5.54

Our three primary research questions/hypotheses for this experiment were:

- (1) Are the aggregate forecasts produced from this proposed judgment aggregation procedure more accurate than the average initial human forecast? H1: **Yes**.

This hypothesis was supported by the data. Across the 200 tasks, the difference between the absolute error of our proposed aggregation procedure and the absolute error of the average initial human forecast **was observed to be negative and significantly different from zero**, using a two-tailed paired t-test of means. **t-statistic: -3.42 Degrees of freedom: 199 p-value: <0.001**

- (2) Are the aggregate forecasts produced from this proposed judgment aggregation procedure more accurate than the average final human forecast? H2: **No**.

This hypothesis was not supported by the data. Across the 200 tasks, the difference between the absolute error of our proposed aggregation procedure and the absolute error of the average final human forecast **was not observed to be significantly different from zero**, using a two-tailed paired t-test of means. **t-statistic: -1.37 Degrees of freedom: 199 p-value: 0.172**

- (3) Are the aggregate forecasts produced from this proposed judgment aggregation procedure more accurate than the AI advice? H3: **No**.

This hypothesis was not supported by the data. Across the 200 tasks, the difference between the absolute error of our proposed aggregation procedure and the absolute error of the AI advice **was observed to be negative and significantly different from zero**, using a two-tailed paired t-test of means.

t-statistic: -1.13 Degrees of freedom: 199 p-value: 0.261